

RESEARCH ARTICLE

Juvenile Incarceration in an Adult Facility and Violent Victimization During Adulthood

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ABSTRACT

Introduction: A growing body of research indicates that being incarcerated in an adult jail or prison as a juvenile can elevate one's exposure to adverse childhood experiences, including violent and sexual victimization and neglect, and may lead to several long-term difficulties. This study investigates the relationship between being confined in an adult jail or prison as a minor and experiencing violent victimization as an adult.

Methods: Data from the National Longitudinal Survey of Youth-1997 cohort were used in the study. Subjects were between 12 and 33 years of age ($N = 8526$; 51% male, 49% female). A parametric survival model was estimated, and the estimates were used to predict the cumulative probability of experiencing a violent victimization from age 18 to 33.

Results: The cumulative probability of experiencing violent victimization by the age of 33 for juveniles who had been detained in an adult facility was approximately 70% higher than for juveniles who have not had contact with the justice system and 22% higher than for arrested juveniles.

Conclusions: Confining juveniles in an adult institution may be a punishment that extends far beyond the sentence, contributing to a unique set of difficulties during adulthood. These findings call into question the practice of bounding juveniles over to the adult court and sentencing them to serve time in adult correctional facilities.

1 | Introduction

Individuals exposed to adverse childhood experiences (ACEs) face substantive disadvantages throughout their life-course, hindering economic and life success and, more importantly, increasing the likelihood of exposure to negative circumstances in the future (Graf et al. 2021). ACEs are experiences captured within three broad categories: (1) abuse, (2) household challenges, and (3) neglect (Felitti et al. 1998). In the context of ACEs, abuse includes physical, emotional, or sexual abuse; household challenges include mental illness amongst family members, incarcerated relatives, interpersonal violence, substance abuse, and divorce; and neglect includes both physical

and emotional neglect. Recent research has expanded the conceptualization of ACEs to include forces within the community, such as bullying, community violence, living in foster care, racism, and neighborhood safety (Cronholm et al. 2015; Merritt et al. 2013).

Exposure to ACEs has been linked to difficulties developing romantic partnerships (Cooper, Adler-Baeder, and McGill 2024), economic hardship (Hardcastle et al. 2018; Liu et al. 2013), involvement in criminal behavior (Basto-Pereira et al. 2022), and victimization (Ports, Ford, and Merrick 2016). This literature, however, often does not focus on the types of environments where children can be exposed to adverse

conditions, instead considering primarily community-based experiences (e.g., Fanslow et al. 2021), and ignoring that children can be exposed to adverse conditions in other environments. One environment known for exposing juveniles¹ to ACEs is jails and prisons that house adults (Ahlin 2020).

Juveniles incarcerated in adult institutions are exposed to negative circumstances that can largely be considered ACEs (Ahlin 2020; Kolivoski and Shook 2016; Murrie et al. 2009). In particular, research suggests that adolescents serving time in adult jails and prisons face a heightened risk of physical and sexual victimization and do not receive the necessary services for proper development during this key period of their life-course (Arya 2007; Canlione and Abrams 2021). Moreover, aligned with the expanded definition of ACEs, juveniles incarcerated in adult jails and prisons will likely be exposed to violence, bullying, and racism during their periods of confinement (Arya 2007; Blazak 2009). Qualitative interviews and self-report surveys suggest that more than half of juveniles incarcerated in adult facilities experience bullying or physical assault, with some individuals reporting multiple victimizations during their period of incarceration (e.g., Heide 2019; Rassouli 2008). The existing literature suggests that the majority of physical victimization is violent in nature, coinciding with a large number of juveniles being violently threatened during their period of incarceration in an adult facility (Arya 2007). In addition, youths incarcerated in adult facilities are likely to be sexualized, as well as experience sexual assault and sexual abuse (Ahlin and Hummer 2019).

The perpetrators are most commonly incarcerated adults, but some adolescents have reported being sexually victimized by security staff working at the adult institution (Ahlin and Hummer 2019). While experiencing physical and sexual abuse is common amongst juveniles incarcerated in adult facilities, juveniles in adult facilities also appear to witness physical and sexual abuse at heightened rates when compared to their counterparts incarcerated in juvenile facilities (Rassouli 2008). Juveniles may be placed in segregation, at times, as a means of protection; however, increased time in isolation may also be considered an ACE as it has been associated with a greater number of mental health diagnoses (Valentine, Restivo, and Wright 2019). Incarcerated adolescents in adult facilities also experience neglect, as these environments typically isolate juveniles for long periods of time and provide them with limited opportunities for social interactions and, in some instances, might not provide youth with basic health care and educational needs (forms of medical neglect and educational neglect; Deitch et al. 2009; Levitt 2010; Ng et al. 2012; Kolivoski and Shook 2016).

Being incarcerated within an adult facility as a juvenile has negative effects on an individual's developmental trajectory into adulthood (Cooper, Adler-Baeder, and McGill 2024). A growing body of literature has documented these effects, suggesting that incarceration in an adult facility is associated with a heightened risk of early child-rearing (Silver, Brookstein, and D'Amato 2023), mental health symptoms (Semenza et al. 2024), unemployment, and lower earnings (Silver, Semenza, and Wooldredge 2024), and future incarceration (Silver, Vaidya, and Wooldredge 2024) during adulthood. These findings coincide with evidence suggesting that individuals who were incarcerated in adult facilities during adolescence have a 33% increased

risk of early adulthood mortality (Silver, Semenza, and Nedelec 2023). Qualitatively, juveniles incarcerated in adult facilities detail how the traumas of their confinement could have contributed to their mental health symptoms and, in turn, substantive disadvantages upon reentry into the community (Rassouli 2008). However, gaps still exist when discussing the consequences of incarcerating adolescents in adult facilities. One particularly salient gap is the potential effects of juvenile incarceration in an adult facility on violent victimization during adulthood.

ACEs, such as, experiencing physical abuse, sexual abuse, neglect, bullying, violence, and racism during adolescence are associated with an increased risk of experiencing violent victimization during adulthood (e.g., Connolly 2020; Fanslow et al. 2021; Ports, Ford, and Merrick 2016).

For instance, Ports et al. (2016) demonstrated that as ACE scores increased, so did the likelihood of experiencing sexual victimization in adulthood amongst a sample of 7272 individuals. Moreover, each ACE—sexual abuse, emotional abuse, psychological abuse, household mental illness, incarcerated household member, emotional neglect, physical neglect, household physical violence, household substance abuse, and parental separation/divorce—was associated with an increased risk of sexual victimization in adulthood. These findings coincide with the literature focused on broader violent victimization, which suggests that ACEs are associated with a heightened likelihood of experiencing a physical assault, burglary, or other forms of violent victimization during adulthood (e.g., Connolly 2020; Voith, Anderson, and Cahill 2020).

Theoretically, exposure to ACEs alters the developmental trajectory of youth and, more importantly, increases their likelihood of being exposed to individuals or situations where they could become victimized (Cronholm et al. 2015; Merritt et al. 2013; Finkelhor and Asdigian 1996). The link between juvenile confinement in adult institutions and adulthood victimization could also exist due to strain within an individual's life during reentry and poor social bonds to the community (Iratzoqui 2019, 2018). However, theory suggests that the routine activities of juveniles previously incarcerated in adult facilities might increase exposure to offenders (Finkelhor and Asdigian 1996). Furthermore, these juveniles also might lack appropriate guardians when reentering the community (e.g., family members) and appear more vulnerable to motivated offenders (Finkelhor and Asdigian 1996).

With adolescents being exposed to ACEs during periods of confinement in adult institutions (e.g., Kupchik 2007) and the line of inquiry demonstrating an association between ACEs and violent victimization during adulthood (e.g., Voith, Anderson, and Cahill 2020), it is reasonable to hypothesize that periods of confinement in adult institutions (jail or prison) as a juvenile could be associated with an increased risk of violent victimization during adulthood. As such, this study aims to answer the following research question: is confinement in an adult institution associated with violent victimization during adulthood? We hypothesize that experiencing a period of confinement in an adult institution as a juvenile is associated with an increased risk of violent victimization during adulthood. Moreover, the association should exist when statistically

controlling for perceived confounders, including legal system covariates, parental indicators, demographics, and other measures captured during adolescence (12–17).

2 | Methods

2.1 | Sample

The current study relied on data from the National Longitudinal Survey of Youth-1997 (NLSY-97) cohort, a publicly available data set collected and archived by the Bureau of Labor Statistics (Moore et al. 2000). This research was exempt from seeking IRB approval as the NLSY-97 is a publicly available deidentified data set. Consent for participation in the NLSY-97 was obtained by NORC at the University of Chicago when enrolling the cohort of individuals who participated in the survey. Through the implementation of a stratified sampling procedure, 9,907 individuals were identified as eligible in households across the United States, and the NLSY-97 enrolled a nationally representative sample of 8,984 individuals born between January 1, 1980 and December 31, 1984, to participate in an initial survey conducted in 1997. Respondents were asked to complete yearly interviews from 1997 until 2011 and then interviews every other year from 2013 to 2021 (20 interviews in total). During the initial interviews, the respondents ranged in age from 12 to 18 years old, resulting in the NLSY-97 research team following the respondents until the age of 42. More information on the NLSY-97 can be found in Moore et al. (2000) or online at <https://www.bls.gov/nls/nlsy97.htm>. The data that support the findings of this study are openly available through the US Bureau of Labor Statistics at <https://www.nlsinfo.org/investigator/pages/home> (see also <https://www.nlsinfo.org/content/cohorts/nlsy97>).

The analytical timeframe for the current study ranged from 1997 to 2013 or when the respondents were between 12 and 33 years of age, as the respondents were last asked about the age at which they experienced a violent victimization during the 2013 interview. The analytical sample for the current study was any individual under the age of 18 during the initial interview (1997-interview; $N = 8961$, > 99% of the NLSY-97 sample) to ensure that all the covariates were measured before the measurement of the outcome variable (violent victimization 18–33). The analytical sample was further limited to only individuals who responded to the violent victimization question at least once across the 2002, 2007, 2008, 2009, and 2013 interviews ($N = 8526$ individuals, 95% of the original sample). A missing case analysis was conducted and suggested that missing information was random across all the covariates, permitting the implementation of random forests multiple imputation to reduce the amount of missing information across the covariates in the current study.² This missing case analysis is provided at: https://anonymous.4open.science/r/JIAF_Violent-Victimization-36D4.

2.2 | Measures

2.2.1 | Violent Victimization 18–33 Years of Age

For a survival model, one must identify who experienced the outcome of interest (i.e., violent victimization after the age of

18) and the time until they experienced the outcome of interest or were censored by data collection procedures. In a survival model, one needs to account for when data collection procedures were censored, as individuals could experience the outcome of interest after data collection concludes. Accounting for the right censorship when estimating the model removes problems associated with censorship bias in traditional binary logistic regression models. This permits the estimation of the survival function, identifying the probability of an individual surviving with each passing period. In the current study, two measures were created to permit the estimation of a survival analysis: (1) if an individual experienced a violent victimization after the age of 18 (“0” = no, “1” = yes) and (2) the number of years until an individual experienced their first recorded violent victimization (“1” represents victimized at the age of 18; 16 represents victimized at the age of 33). For those individuals that did not experience a violent victimization, the value recorded on the number of years until the individual experienced their first recorded violent victimization was their age in 2013 minus 17 (e.g., 16 years), which captures when data collection was concluded—censored—for the corresponding individual.

The responses to two questions in the NLSY-97 were used to operationalize these measures. The respondents answered the question, “In the last 5 years, have you been the victim of a violent crime, for example, physical or sexual assault, robbery, or arson?” Then, for those individuals who did experience a violent victimization, the value recorded on the number of years until the individual experienced their first recorded violent victimization was derived from the question, “How old were you when you were the victim of a violent crime? If you have been victimized more than once, please tell me your age the last time you were a victim,” to which the respondents were required to provide an age at which the victimization occurred. The respondents were only asked to record a response to these questions in 2002, 2007, 2008, 2009, and 2013, making it difficult to determine if the respondents experienced a violent victimization before 1998 or before the age of 18.³ To calculate the number of years until the individual experienced their first recorded violent victimization after the age of 18, 17 was subtracted from the age recorded in response to each question, and the lowest value was kept. For example, if a respondent experienced a violent victimization at the age of 25, they received a value of “8” on the years until the individual experienced their first recorded violent victimization.

2.2.2 | Incarceration in an Adult Facility as a Juvenile (12–17 Years of Age)

The NLSY-97 research team asked the respondents to report the start and end date of each period of confinement they experienced, which was then used to create a monthly array of indicators identifying if the respondent was incarcerated in an adult jail or prison facility from 1992 to 2019 (Silver, Semenza, and Wooldredge 2024). A dichotomous indicator was created to identify the respondents who were incarcerated in an adult jail or prison facility before the age of 18, using the age of the respondents each year based on their age in the months recorded in 1997. A value of “1” on this indicator identifies that

the respondent was incarcerated in an adult correctional facility as a juvenile (<18 years of age), while a value of “0” on this indicator identifies that the respondent was not incarcerated in an adult correctional facility before the age of 18 (e.g., no contact with the criminal justice system or arrested but not incarcerated in an adult facility). This operationalization aligns with the legal definition of a juvenile, which in many states are individuals under the age of 18.

2.2.3 | Legal System Covariates (12–17 Years of Age)

Two covariates were included to adjust for the behavioral tendencies of the respondents and other factors that could potentially confound the association between experiencing a period of confinement in an adult institution and violent victimization during adulthood. First, a dichotomous measure was created to indicate if a respondent had been arrested before 18 but not incarcerated in a facility (“1” = yes; reference category is no contact with the criminal legal system). Second, an indicator was created to identify if a respondent experienced a period of confinement in a juvenile correctional facility before 18 (“1” = yes; reference category is not incarcerated in a juvenile correctional facility before the age of 18).

2.2.4 | Parental Indicators

Four parental indicators were included in the statistical analyses to adjust for their potential influence on the association of interest and, more importantly, reduce the biases that could arise from adverse household environments influencing both confinement in an adult correctional facility as a juvenile and future violent victimization. First, parental incarceration before the age of 16 was a dichotomous measure created using the responses to the following question: “As far as you know, did your [father or mother] ever serve a prison sentence for a conviction before your 16th birthday? Please include any time she may have served before you were born.” Responses to this question were recorded as “0” = no and “1” = yes. Second, the biological mother’s educational attainment was coded as the highest grade level completed by the biological mother as of the 1997 interview (ranging between 1st and 20th grade). Third, the biological father’s educational attainment was coded as the highest grade level completed by the biological father as of the 1997 interview (ranging between 1st and 20th grade). Finally, parental health captures the self-reported health of the parent who completed the parent interview. Scores on this item ranged from 1 to 5, where higher values represent better health of the parent.

2.2.5 | Demographics

The demographic characteristics of the respondents were introduced into the statistical analysis to adjust for potential differences in the key variables. The ethnicity of the respondent was recorded as Hispanic (“1” = yes; reference category is non-Hispanic). The racial heritage of the respondent was recorded as Black (“1” = yes) or other race (“1” = yes, including Asian,

Pacific Islander, Native American, etc.), with White serving as the reference category. The biological sex of the respondent was recorded as male (“1” = yes), with female serving as the reference category.

2.2.6 | Covariates Measured in 1997

Thirteen additional covariates were introduced into the statistical models to adjust for the potential effects of household, peer, and school experiences on the association of interest. Adverse experiences are dichotomous indicators recording the parents’ responses to the question “Sometimes children go through hard times. For example, they live in a place without water or electricity, or in a homeless shelter. To the best of your knowledge, has [this youth] ever experienced such hard times?” with “1” representing yes and “0” representing no. Household net worth was the parental report of the total assets of the household the participant resided in during the 1997 interview minus the total debts of the household. Percent of peers who smoke, drink, were in a gang, used illegal drugs, or had sex were derived from a series of items asking the respondent to record the percentage of peers who engaged in each of these behaviors on a 1–5 scale. A value of “1” equals less than 10%, “2” equals about 25%, “3” equals about 50%, “4” equals about 75%, and “5” equals more than 90%. Threatened at school and fought at school are self-reported measures capturing the number of times the respondent had these experiences, where higher values indicate that the individual was threatened or fought more often.

Late for school and days absent from school represent the total number of days the individual was late to or absent from school during the most recent term. Higher values indicate a greater frequency of these attendance issues. Teachers are good and discipline is fair are ordinal measures capturing the respondents’ perceptions of the teachers within their school and school discipline. These items were scored on a 1–4 scale, where “1” = strongly disagree, “2” = disagree, “3” = agree, and “4” = strongly agree.

Briefly, time varying covariates were not included in the statistical models as the primary interest was estimating the *total effects* of youth incarceration in adult facilities on adulthood victimization, which includes the *direct effects* as well as the *indirect effects* through mediating mechanisms such as time spent in facilities as an adult. The process of estimating total effects, as opposed to direct and indirect effects is detailed by Judea Pearl (2009) and Robert Kline (2023). As https://github.com/ianasilver/JIAF_Violent-Victimization such, the current study was interested in estimating the *total effects* (*direct effects* plus *indirect effects*) of youth incarceration in adult facilities on employment and income after adjusting for potential confounders and adolescent mediators. We believe the *total effects* of youth incarceration in adult facilities, rather than isolating the *direct effects*, provides a more complete understanding of the effects on adulthood victimization. The analytical strategy use for the current study provides the best approach to test these total effects, because including time-varying covariates after the age of 18 could potentially mean that we are estimating the *total effects* for some individuals and the *direct effects* for other individuals.

2.3 | Analytical Strategy

A five-part analytical strategy was implemented to answer the identified research question and evaluate the association between incarceration in an adult facility as a juvenile and the time until experiencing a violent victimization after the age of 18. First, a missing case analysis was conducted (see https://github.com/ianasilver/JIAF_Violent-Victimization). The results of the missing case analysis suggested that the data was missing at random and permitted the employment of multiple imputation using chained equations with a random forest estimator to address missing information across the covariates of interest (Little and Rubin 2019; Shah et al. 2014).⁴ Second, post-imputation descriptive statistics were produced.

Third, a parametric survival model was estimated regressing the years until an individual experienced a violent victimization after the age of 18 on juvenile incarceration in an adult correctional facility and the key covariates, the parental indicators, demographics, and the covariates measured in 1997. A parametric survival model was preferred in the current instance as the change in the likelihood of experiencing a violent victimization from year to year was not proportional across juveniles incarcerated in an adult correctional facility and juveniles not incarcerated in an adult correctional facility, violating the proportional hazards assumption (Hess 1995; Shepherd 2008). The parametric survival model was estimated using a lognormal distribution, which was determined after comparatively evaluating the distributional qualities of the survival function for years until an individual experienced a violent victimization after the age of 18. In particular, years until violent victimization was subjected to six univariate survival functions with unique distributional assumptions—lognormal, exponential, Weibull, Gaussian, logistic, and loglogistic—and the Akaike Information Criterion (AIC)s for each survival function were compared (Wagenmakers and Farrell 2004). The distributional assumptions resulting in the survival function with the lowest AIC were selected: *lognormal*.

Longer analytical timeframes can result in higher standard errors (due to more substantive increases in the probability of the outcome when moving from one observation to the next) and a reduced likelihood of detecting statistical significance (Hosmer, Lemeshow, and May 2008). Additionally, using years can create a smoother curve permitting it to satisfy the assumptions of survival analysis more readily (Clark et al. 2003). In a more practical sense, a more fine-grained analysis could provide a better understanding of when intervention should occur for these youth.

Fourth, the estimates from the parametric survival model were used to predict the cumulative probability of experiencing a violent victimization from age 18–33 for (1) youth incarcerated in adult facilities, (2) youth arrested but not incarcerated in adult facilities, and (3) youth that experienced no contact with the criminal justice system (Crowther and Lambert 2014). These cumulative probabilities were then plotted to observe the growth in the likelihood of experiencing violent victimization as individuals aged within the three groups. Finally, to ensure that the results presented in the primary text were consistent across analytical techniques, the models were replicated using a

matched subsample, removing individuals incarcerated in a facility during the age of 18, and estimating the model with and without legal system covariates. The matched subsample was created through the employment of 4–1 optimal propensity score matching without replacement to match juveniles incarcerated in adult facilities (treatment cases) to similar individuals that never experienced a period of confinement in an adult facility under the age of 18 (control cases; Chen et al. 2022; Rosenbaum 1989). The results of these replications are discussed in the supplemental results section and presented in Supporting Information S1: Appendices A–C.

3 | Results

3.1 | Primary Results

Table 1 provides the post-imputation descriptive statistics for the analytical sample. Focusing on the variables of interest, it can be observed that approximately 12.1% of the sample experienced a violent victimization after the age of 18 ($N=1028$) and, on average, they were victimized between the ages of 24 and 25. Given the standard deviation ($SD=3.93$), the majority of the individuals that did experience a violent victimization event experienced it between 20 and 29 years of age. Of the analytical sample, approximately 1.2% or 106 individuals were incarcerated in adult facilities before the age of 18, with 184 individuals (2.2%) experiencing a period of confinement in a juvenile facility before the age of 18, and 1446 individuals (17.0%) experiencing an arrest without any incarceration before the age of 18.

Table 2 provides the results of the parametric survival model regressing the years until an individual experienced a violent victimization after the age of 18 on juvenile incarceration in an adult institution, the key covariates, parental indicators, demographics, and covariates measured in 1997. Supporting Information S1: Appendix D provides the multicollinearity evaluation for the parametric survival model. Evident by the tolerance and the variance inflation factor (VIF), the slope coefficients and standard errors should not become biased due to violations of the multicollinearity assumption (Thompson et al. 2017). That is, none of the tolerance values were below 0.400 and none of the VIF values were above 2.500. Focusing on the key findings, it can be observed that being incarcerated in an adult institution before the age of 18 is associated with a decreased number of years until experiencing a violent victimization. That is, time ratio ($b=-0.690$; $p=0.010$; $TR=0.501$) suggests that juveniles incarcerated in adult institutions experience a 50% reduction in the number of years until experiencing a violent victimization ($p=0.010$). Independent of being incarcerated in an adult institution, the findings suggest that being arrested as a juvenile was associated with experiencing a 30% reduction in the number of years until violent victimization ($b=-0.402$; $p=0.000$; $TR=0.669$). Overall, these findings suggest that contact with the criminal legal system as a juvenile is associated with an increased risk of violent victimization, with juvenile incarceration in adult institutions having a particularly salient reduction in the number of years until experiencing a violent victimization.

Figure 1 presents the plotted cumulative predicted probability of experiencing violent victimization from age 18 to 33 for

TABLE 1 | Descriptive statistics for the analytical sample (after imputation).

	Mean (%)	SD	Min	Max
Dependent variables				
Violent victimization after 18	(12.1)	0.33	0.00	1.00
Number of years until violent victimization after 18±	5.97	3.93	1.00	16.00
Treatment (12–17 years of age)				
Incarcerated in adult institution	(1.2)	0.11	0.00	1.00
Legal system covariates (12–17 years of age)				
Arrested	(17.0)	0.38	0.00	1.00
Incarcerated in juvenile correctional facility	(2.2)	0.15	0.00	1.00
Parental indicators				
Parental incarceration before 16	(6.6)	0.25	0.00	1.00
Bio-mother educational attainment	12.41	2.93	1.00	20.00
Bio-father educational attainment	12.34	3.23	1.00	20.00
Parental health in 1997	3.67	1.10	1.00	5.00
Demographics				
Hispanic	(21.4)	0.41	0.00	1.00
Black	(27.0)	0.44	0.00	1.00
Other race	(14.6)	0.35	0.00	1.00
Male	(51.9)	0.50	0.00	1.00
Sample	(25.3)	0.43	0.00	1.00
Covariates measured in 1997				
Adverse experiences	(5.1)	0.22	0.00	1.00
Household net worth	91,856.69	140,256.89	−935,251.00	600,000.00
% Peers smoke	2.61	1.28	1.00	5.00
% Peers drink	2.12	1.24	1.00	5.00
% Peers in gang	1.60	0.99	1.00	5.00
% Peers who use illegal drugs	2.30	1.31	1.00	5.00
% Peers had sex	2.47	1.33	1.00	5.00
Threatened at school	0.99	5.30	0.00	99.00
Fought at school	0.41	1.80	0.00	50.00
Late for school	2.41	6.99	0.00	99.00
Days absent from school	4.92	7.92	0.00	200.00
Teachers are good	3.02	0.62	1.00	4.00
Discipline is fair	2.81	0.78	1.00	4.00
N			8526	

Note: “±” indicates that the descriptive statistics only derived from those victimized. All non-victimized individuals received a value representing the difference between their age during the 2013 interview and 18 (e.g., age in 2013–2018).

individuals who experienced a period of incarceration in an adult facility before 18, were arrested before 18, and had no contact with the criminal legal system before 18. The findings further support the results presented in Table 2, suggesting that at every age period, juveniles who were incarcerated in adult correctional facilities have a heightened probability of experiencing a violent victimization. In particular, by the age of 33, 17 of every 100 juveniles incarcerated in adult facilities will experience a violent victimization, which is substantively higher than juveniles arrested before the age of 18 (14 of every 100) and

juveniles who never had contact with the criminal legal system (10 of every 100).

3.2 | Supplemental Results

In addition to the primary model, two supplemental models were estimated to assess the validity of the findings presented in the primary text. First, the parametric survival model was replicated using a subsample of individuals incarcerated in

TABLE 2 | Parametric survival models predicting years after 18 until violent victimization with juvenile incarceration in adult correctional facility.

	DV: Number of years until violent victimization (after 18)					
	<i>b</i>	SE	<i>p</i> value	TR	TR L 95% CI	TR U 95% CI
Treatment (12–17 years of age)						
Incarcerated in adult institution	−0.690	0.268	0.010	0.501	0.297	0.847
Legal system covariates (12–17 years of age)						
Arrested	−0.402	0.096	0.000	0.669	0.554	0.808
Incarcerated in juvenile correctional facility	0.094	0.241	0.698	1.098	0.685	1.762
Parental indicators						
Parental incarceration before 16	−0.672	0.126	0.000	0.511	0.399	0.654
Bio-mother educational attainment	−0.003	0.016	0.837	0.997	0.967	1.028
Bio-father educational attainment	−0.044	0.014	0.002	0.957	0.931	0.984
Parental health in 1997	0.002	0.035	0.946	1.002	0.937	1.073
Demographics						
Hispanic	−0.138	0.123	0.259	0.871	0.685	1.107
Black	0.106	0.106	0.318	1.111	0.903	1.368
Other race	0.166	0.122	0.175	1.180	0.929	1.499
Male	0.136	0.073	0.064	1.145	0.992	1.322
Sample	−0.093	0.101	0.359	0.912	0.748	1.111
Covariates measured in 1997						
Adverse experiences	−0.414	0.147	0.005	0.661	0.495	0.882
Household net worth	0.000	0.000	0.489	1.000	1.000	1.000
% Peers smoke	−0.013	0.039	0.741	0.987	0.914	1.066
% Peers drink	0.019	0.042	0.647	1.019	0.939	1.107
% Peers in gang	−0.093	0.039	0.016	0.911	0.844	0.983
% Peers who use illegal drugs	0.003	0.040	0.946	1.003	0.927	1.085
% Peers had sex	−0.043	0.037	0.246	0.958	0.891	1.030
Threatened at school	−0.016	0.006	0.006	0.984	0.973	0.995
Fought at school	0.005	0.016	0.779	1.005	0.973	1.037
Late for school	−0.006	0.004	0.159	0.994	0.985	1.002
Days absent from school	−0.007	0.004	0.100	0.993	0.986	1.001
Teachers are good	0.010	0.061	0.872	1.010	0.897	1.137
Discipline is fair	0.073	0.049	0.133	1.076	0.978	1.184
Log (scale)	0.718	0.021	< 0.001			
<i>N</i>				8526		

adult facilities matched to individuals who were not incarcerated in adult facilities (see Supporting Information S1: Appendix A). The matching procedure was executed using optimal matching with a 4 to 1 control case to treatment case ratio without replacement. This matching procedure achieved a high degree of balance across the treatment and control groups, permitting the estimation of the post-matching parametric survival model. Focusing on the results of that model, it appeared that juveniles incarcerated in adult institutions experienced a violent victimization approximately 7–8 months earlier than juveniles who have never had contact with the

criminal legal system, which corresponds to a 48% reduction in the number of years until violent victimization ($p = 0.027$; Supporting Information S1: Table A2). Second, the parametric survival model was replicated removing individuals who were incarcerated at the age of 18 to establish a well-defined temporal gap between juveniles incarcerated in adult institutions and violent victimization after the age of 18. As evident by the findings, the estimates were largely identical to the estimates presented in the main text, suggesting that juveniles incarcerated in adult institutions experienced a violent victimization approximately 57% earlier than juveniles who have never had

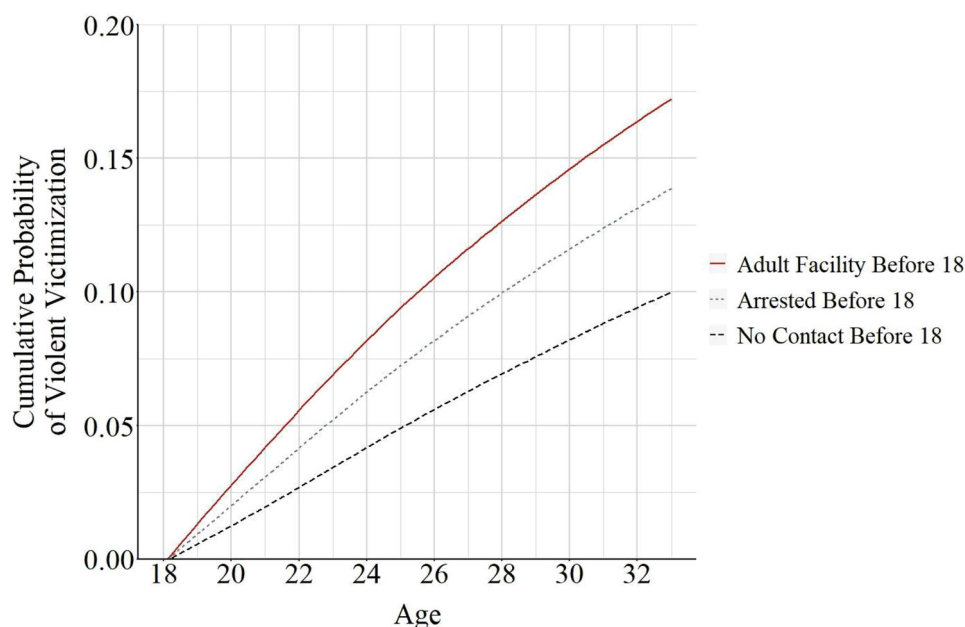


FIGURE 1 | Predicted Cumulative Probability of Experiencing a Violent Victimization After 18th Birthday. Predicted cumulative probability is based on the model results presented in Table 2. “Adult Facility Before 18” represents juveniles incarcerated in adult facilities only; “Arrested Before 18” represents juveniles arrested but not incarcerated in adult or juvenile facilities before the age of 18; “No Contact Before 18” represents juveniles that did not encounter the criminal legal system before the age of 18.

contact with the criminal legal system ($p = 0.013$; Supporting Information S1: Table B1).

Finally, a series of models with and without the legal system covariates were estimated (see Supporting Information S1: Appendix C). The results of these supplemental models were consistent with the effects observed in the primary text. In particular, the results suggested that individuals incarcerated in adult institutions before the age of 18 experienced between a 42% reduction and a 50% reduction in the number of years until experiencing a violent victimization. All of these effects were observed at $p < 0.05$, further supporting the conclusion that individuals incarcerated in adult institutions before the age of 18 experienced violent victimization earlier and more frequently in adulthood.

4 | Discussion

Three findings from the current study should be highlighted. The first finding of the study was that individuals who experienced violent victimization in adulthood were, on average, between the ages of 24 and 25, or 5.97 years after the age of 18. This suggests that most individuals who experienced violent victimization after turning 18 did so during early adulthood. This result is consistent with national estimates, which illustrate that individuals ages 18–24 and 25–34 have the highest prevalence of violent victimization (Thompson and Tapp 2022). Existing theoretical perspectives suggest that the age victimization association could be due to a multitude of factors, including, but not limited to, a lack of guardianship while establishing oneself and increased risk-taking behaviors (Fox and Bouffard 2015).

The second finding of the study was that youths arrested before the age of 18 had an increased likelihood of experiencing violent

victimization earlier during adulthood than youths who never experienced contact with the criminal justice system. It appears juveniles arrested before the age of 18 are more likely to experience a violent victimization 4–5 months earlier than those not involved in the system. This result suggests that the adverse experience of being arrested can potentially influence future violent victimization, placing justice-involved youth at a heightened risk for adverse experiences during early adulthood. These findings, however, are not surprising as scholarship has consistently found a link between adolescent offending and victimization during adolescence/early adulthood (Jennings, Piquero, and Reingle 2012). Offending and victimization are highly correlated and, in some instances, reciprocal in nature, suggesting that early life-course arrests could lead to future violent victimization through a variety of mechanisms (Jennings, Piquero, and Reingle 2012). These mechanisms include, but are not limited to poor guardianship, increased involvement in deviant behaviors (e.g., drug use), and acquired psychological symptoms (Silver and Kelsay 2022).

The third finding of the study was that youths incarcerated in adult facilities before the age of 18 had an increased likelihood of experiencing violent victimization earlier during adulthood than youths who never experienced contact with the criminal justice system after adjusting for the effects of being arrested on adulthood violent victimization. Specifically, juveniles who served time in an adult institution were likely to experience victimization 8–9 months earlier than those not criminally-involved. These findings suggest that the confinement of a juvenile in an adult institution might be considered a risk factor for violent victimization during early adulthood (18–33 years of age), aligning with the prior literature suggesting that the experiences of juveniles in adult institutions—for example, ACEs—or the routine activities of these individuals after

incarceration might contribute to negative outcomes during adulthood (e.g., Semenza Silver and Jackson 2024; Silver, Semenza, and Nedelec 2023). There could be a variety of mechanisms driving the observed association, none of which is more prominent than the increased likelihood of experiencing ACEs during incarceration in adult facilities. As detailed, juveniles experience or witness physical and sexual assaults and experience various forms of neglect at a heightened rate when incarcerated in adult facilities. Experiencing ACEs during a critical period in one's development has been shown to have a lasting effect, contributing to a variety of negative outcomes during adulthood, including violent victimization (Arya 2007). In particular, experiencing these types of ACEs from 12 to 18 can alter one's biological and social developmental trajectory (Kerker et al. 2015; Navalta, McGee, and Underwood 2018), exposing them circumstance that increase one's risk of violent victimization. Furthermore, exposure to physical and sexual abuse can alter one's brain structure and brain development (including the prefrontal cortex and amygdala), altering one's decision-making process during early adulthood, and leading to decisions that could increase violent victimization (Navalta, McGee, and Underwood 2018). As such, if the incarceration of a juvenile in adult facilities does result in a heightened risk of experiencing ACEs, the lasting effects of those carceral experiences could be an increased risk of violent victimization during early adulthood.

The current study is not without limitations, seven of which should be highlighted. First, due to data collection procedures, we could not operationalize violent victimization before the age of 18. Second, the respondents were not required to provide the location of the violent victimization, making it difficult to determine if the violent victimization experienced during early adulthood occurred in the community or in a carceral setting. While examining the effects of juvenile incarceration in adult facilities on early adulthood victimization regardless of the setting was the focus of the current study, future research should determine if the magnitude of the association differs by the setting in which the victimization occurred. Third, due to data limitations, we were unable to operationalize ACEs experienced during incarceration, removing our ability to directly test the mechanisms driving this association. Future research should consider testing if (1) juveniles experience heightened ACEs during periods of confinement in adult institutions, and (2) those ACEs experienced contribute to violent victimization during early adulthood. Fourth, the NLSY97 data set was initially collected on the 1980–1984 birth cohort and while this does provide the ability to follow respondents into adulthood, the results might be different in a more contemporary data set. Future scholarship should replicate the current study with a more contemporary sample. Fifth, due to data collection limitations, time until victimization could only be operationalized in years since an individual's 18th birthday. Future research should operationalize time until victimization in months or days to observe if a similar pattern of findings is observed and additional insights can be gained through greater precision. Sixth, the current study did not examine the mechanisms mediating the association between juvenile incarceration in adult facilities and future victimization. Future research should extend the current study to examine the mechanisms that mediate this association. Finally, due to the NLSY97 research

team requesting the respondent's age at the violent victimization event—and not the month/year of the violent victimization event—we are unable to further test the findings using months as the unit of analysis (as opposed to years). Future research should reexamine this research question using months as the unit of analysis.

The findings of the current study suggest that the potential long term impact of the adult carceral setting on youth should be considered when sanctioning adjudicated juveniles. Specifically, this manuscript contributes to a growing body of literature that suggests that the confinement of juveniles in adult institutions could result in negative life-course outcomes beyond the individual's behavioral tendencies (e.g., Silver, Semenza, and Nedelec 2023). Of particular importance, victimization during early adulthood and adulthood could contribute to the heightened risk of mortality and mental health diagnoses amongst juveniles previously incarcerated in adult facilities (Semenza, Silver, and Jackson 2024). The confinement of a juvenile in an adult institution might be considered “cruel and unusual,” contributing to a unique set of difficulties during early adulthood and adulthood. This includes a heightened cumulative probability of experiencing violent victimization by the age 33, approximately 70% higher than juveniles who have not had contact with the criminal legal system and 22% higher than arrested juveniles. If the growing body of literature illustrates additional harms of confining juveniles in adult officials, it may become increasingly difficult to observe confinement in adult institutions as a just punishment for adjudicated juveniles.

Acknowledgments

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in NLSY repository at <https://www.nlsinfo.org/investigator/pages/home>.

Endnotes

¹Juveniles, children, and adolescents are used interchangeably throughout the manuscript and refer to individuals under the age of 18—the legal definition of a juvenile across most states in the United States (Rassouli 2008). Juveniles can become incarcerated in adult institutions by being automatically bound over to the adult court during adjudication or by the prosecutor and judge deciding if the crime is serious enough for the juvenile to be tried in adult courts. Juveniles as young as 12-years of age—in some states—could be punished to serve time in adult institutions at the discretion of the prosecutor and judge.

²The primary dependent and independent variables was non-missing for all individuals included in the analytical sample.

³The respondents ranged in age from 17 to 23 in 2002 and 28 to 33 in 2013.

⁴No missing information existed in the primary independent variable (incarcerated in an adult facility as a juvenile) and the primary dependent variable (violent victimization after the age of 18).

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.